

AI-on-RAN for Cyber Defense: An XAI-LLM Framework for Interpretable Anomaly Detection

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Abstract—Next generation Radio Access Networks (RANs) introduce programmability, intelligence, and near real-time control through intelligent controllers, enabling enhanced security within the RAN and across broader 5G/6G infrastructures. This paper presents a comprehensive survey highlighting opportunities, challenges, and research gaps for Large Language Model (LLM)-assisted explainable (XAI) Intrusion Detection Systems (IDS) in future RAN environments. Motivated by this, we propose an LLM interpretable anomaly detection system leveraging multivariate time series Key Performance Measures (KPMs), extracted from E2 nodes, within the Near Real-Time RAN Intelligent Controller (Near-RT RIC). A sequence classification model is trained to identify malicious User Equipment (UE) behavior based on these KPMs. To enhance transparency, we apply post-hoc local explainability methods such as LIME and SHAP to interpret individual predictions. Furthermore, LLMs are employed to convert technical explanations into natural-language insights accessible to non-expert users. Experimental results on real 5G network KPMs demonstrate that our framework achieves high detection accuracy (macro F1-score > 0.96) while delivering actionable and interpretable outputs.

Index Terms—XAI, LLMs, AI RAN, DDoS, near-RT-RIC, network security.

I. INTRODUCTION

THE Open Radio Access Network (RAN) paradigm has significantly reshaped the modern wireless network architecture by introducing disaggregated, interoperable, and programmable components into the RAN [1], [2]. A key enabler of this flexibility is the introduction of intelligent RAN controllers, specifically the Near-Real-Time RIC (Near-RT RIC) and the Non-Real-Time RIC (Non-RT RIC), which respectively support the deployment of custom applications known as xApps and rApps [3], [4]. These applications facilitate intelligent control, optimization, and monitoring of radio resources by leveraging

analytics and Machine Learning (ML) across different time scales. This software-driven programmability enhances network adaptability and automation, supporting dynamic network slicing, advanced security mechanisms, resource allocation, and orchestration processes that flexibly adapt to diverse and evolving network demands [5], [6]. Building on these architectural foundations, the AI RAN paradigm has emerged as a natural next step, aiming to integrate AI-driven intelligence across all layers of the RAN. Defined by the AI-RAN Alliance, this paradigm outlines three key development areas: AI-for-RAN, AI-on-RAN, and AI-and-RAN [7], [8], each addressing a distinct aspect of AI integration in the design and operation of RAN systems.

In this context, AI-RAN provides new opportunities to harness AI to improve network security. Both AI-for-RAN and AI-on-RAN enable the development of intelligent, adaptive mechanisms for detecting and mitigating anomalous behavior originating from User Equipment (UE) or network entities. Such mechanisms are increasingly critical as mobile networks face sophisticated and evolving security challenges, where malicious or misbehaving UEs can disrupt control and data plane operations, degrade quality of service, or even impact network stability [9], [10]. Traditional defense mechanisms often reside in the core network, limiting their responsiveness to edge originated anomalies. Detecting abnormal activity earlier, closer to the RAN edge, can substantially reduce reaction time and potential damage. Leveraging telemetry data such as Key Performance Measurements (KPMs) exposed via E2 interfaces [11], [12], [13], the Near-RT RIC can host intelligent anomaly detection xApps that analyze UE behavior in near real time. As a representative example, this paper considers the case of a Distributed Denial-of-Service (DDoS) pattern to illustrate the framework's operation, however, the proposed approach generalizes to other anomaly types and security-related deviations in RAN performance. On key contribution of this work lies in advancing the AI-for-RAN development area by proposing an ML-based framework for anomaly detection using multivariate time-series KPMs collected from UEs. A sequence modeling classifier is trained to identify abnormal or potentially malicious UE behavior based on these temporal features.

Although detection models can achieve strong predictive performance, their decision process remains opaque, which poses challenges for trust, validation, and operational use in security-critical environments [14], [15]. This is where the AI-on-RAN

Received 17 July 2025; revised 22 October 2025; accepted 2 November 2025. Date of publication 6 November 2025; date of current version 11 December 2025. This work was supported in part by the ORAN-TWIN-X Project of CHEDDAR: Communications Hub for Empowering Distributed Cloud Computing Applications and Research, funded by the U.K. Engineering and Physical Sciences Research Council under Grant EP/Y037421/1 and Grant EP/X040518/1, in part by the Mobile oRAN for highly Dense Environments (5G MoDE) Project, in part by the TUDOR (Towards Ubiquitous 3D Open Resilient Network) Project, and in part by two major winners of the U.K.'s Department of Science, Innovation and Technology Open Networks Ecosystem Competition. Recommended for acceptance by Dr. Yao Sun. (Corresponding author: Mohammad Shojafar.)

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Digital Object Identifier 10.1109/TNSE.2025.3629983

paradigm plays a key role, by enabling the deployment of auxiliary AI models, including Large Language Models (LLMs) and generative AI tools. Recent advancements in AI-native network architectures [16], [17], [18] exemplify this approach, showcasing the collaborative deployment of LLMs across cloud and edge environments. Such architectures facilitate real-time generative AI services and intelligent network management that can enhance RAN functionality, improve overall network security, and support operator decision-making. Recent research further illustrates this broader trend toward interpretable and automated AI-native 6G RANs, with examples addressing intelligible protocol learning for O-RAN slicing [19], explainable reasoning for radio resource allocation [20], and generative simulation for 5G/6G networks [21].

Existing works either focus solely on detection or explainability without integrating automated reasoning via LLMs, or they do not explicitly address the operational time constraints critical for deployment in Near-RT RIC environments. To address this, our framework first integrates Explainable AI (XAI) techniques to provide local explanations for individual model predictions, highlighting which features contributed to a specific classification. However, raw XAI output, such as feature attribution scores, can be complex and difficult for non-expert personnel to interpret quickly and act upon. To overcome this, these local explanations are passed through an LLM that automatically translates complex technical outputs into human-readable natural language insights, and it leverages its reasoning capabilities (if applicable) to suggest potential mitigation strategies based on detected anomalies. By doing so, the LLM reduces the cognitive burden on operators, accelerates incident response, and moves toward partial security workflow automation, bridging the gap between anomaly detection and mitigation.

Contributions: The main contributions of this paper are summarized as follows:

- 1) We provide a comprehensive survey on LLM and XAI integration for Intrusion Detection Systems (IDS) for RAN security in future networks. We highlight the key challenges and opportunities as well as the existing research gaps in this area.
- 2) We propose an LLM-explainable anomaly detection framework leveraging a sequence modeling approach operating on UE KPM time-series data.
- 3) We integrate local XAI methods (LIME and SHAP) to explain per-instance predictions and enhance interpretability.
- 4) We introduce the use of LLMs to translate technical explanations into natural language, increasing human readability.

The remainder of the paper is organized as follows. Section II reviews related work on intrusion detection, XAI, and LLM in wireless networks. Section III introduces the proposed framework, detailing the system architecture, the classification model, and the interpretability pipeline. Section IV presents the experimental evaluation, which assesses both detection accuracy and explanation quality. Finally, Section V concludes the paper and outlines potential future directions.

II. RELATED WORK

This section reviews related work on IDSs, with a focus on the integration of ML and Deep Learning (DL) techniques, the role of XAI in improving the transparency of IDS decisions, and the emerging use of LLMs for intrusion detection, mitigation, and explainability.

A. Intrusion Detection Systems

IDSs serve as a critical component in the defense of a multitude of networks. Examples include SDN-based IDSs, which leverage centralized control and network programmability for enhanced threat detection [22], [23], [24], IoT-oriented IDSs, which address the unique constraints of low-power heterogeneous devices [25], [26], [27], and IDSs designed for vehicular networks, which address high mobility and real-time communication requirements [28], [29], [30]. Unlike static defenses that rely on predefined rules or known threat signatures, IDSs are designed to dynamically observe network activity and identify potentially harmful patterns. In this subsection, we review key developments in classical, ML-based, and DL-based IDSs, with special attention to their application in mobile networks, 5G / B5G infrastructure, and next-generation RAN architectures.

1) *Classical, ML-Based and DL-Based IDS:* The evolving landscape of cyber threats and the limitations of rule-based detection systems drove the evolution of IDSs. Early systems relied on signature matching to identify known attacks, but such approaches struggled with novel threats [31]. This limitation led to behavior-based and anomaly-driven methods, where deviations from learned normal patterns were flagged as potential intrusions. ML played a central role in enabling this shift by providing data-driven models to analyze complex traffic and user behavior [32]. Multiple surveys in the past few years reviewed IDS approaches, each providing a different emphasis on scope, methods, and challenges. Ahmad et al. [33] provided a systematic review of network IDS (NIDS) utilizing ML and DL approaches. They highlighted recent methodologies, evaluated their strengths and weaknesses, and identified key datasets and evaluation metrics commonly used in the literature. Similarly, Abdulganiyu et al. [34] conducted a systematic review of the literature for signature, anomaly, and hybrid-based IDSs, evaluating their performance metrics, datasets used, and attack detection capabilities. Xu et al. [35] focused specifically on the DL-based IDS lifecycle, emphasizing data collection, log parsing, and intrusion detection methods. They thoroughly examined DL methodologies such as Graph Neural Networks (GNNs), transformers, and recurrent models, highlighting their effectiveness in detecting both known and zero-day vulnerabilities. Khraisat et al. [36] presented an in-depth survey on the application of Federated Learning (FL) in IDS, analyzing architectures and aggregation strategies for privacy-preserving intrusion detection. The authors analyzed the strengths and limitations of FL methodologies, addressing significant issues like data privacy, security, and scalability in IDS. Lastly, He et al. [37] reviewed recent literature on adversarial ML, specifically targeting DL-based NIDS. They categorized adversarial attacks into

white-box and black-box methods, highlighting their specific applications and limitations in the network security context. Additionally, the authors evaluated existing defensive strategies against adversarial attacks, outlined significant challenges, and proposed future research directions to improve the robustness of DL-based NIDS.

2) *Temporal Modeling for IDSs*: In certain intrusion detection scenarios, it is necessary to analyze sequences of temporally ordered data samples rather than isolated instances to accurately identify specific attack patterns. Many behaviors, such as DDoS attacks, become apparent only through temporal dependencies between data points. DL approaches have been shown to be effective in modeling such temporal dependencies for anomaly detection in complex network data [38]. To address this, recent research has incorporated temporal modeling techniques that enable systems to learn from the ordering and evolution of events. This shift has brought increased attention to sequential architectures such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTMs), Gated Recurrent Units (GRUs), and other more cost-efficient variations [39], and more recently, attention-based and transformer models, which offer improved capabilities for capturing long-range dependencies. For example, Dash et al. [40] proposed an optimized LSTM-based framework for anomaly detection, using metaheuristic hyperparameter tuning, to showcase how recurrent models improve detection over conventional ML-based frameworks. Building on these advances, recent studies have shown that integrating attention mechanisms with RNN, CNN, and hybrid CNN-BiLSTM architectures can further enhance temporal feature learning and improve intrusion detection performance [41], [42]. Temporal Convolutional Networks (TCNs) have also been introduced as an effective alternative to modeling long-range dependencies in sequential network traffic [43]. Lastly, transformer-based architectures have enabled flexible sequence modeling for network intrusion detection and have been applied to optimize the trade-off between detection accuracy and timely detection in real-time settings [44], [45].

3) *IDSs in 5G, B5G, and Next-Generation RAN*: The evolution of mobile networks toward 5G, beyond 5G (B5G), and next-generation RANs, including paradigms such as Open RAN and AI-RAN, has introduced substantial new security challenges [46], [47]. Advanced infrastructures introduce new components, interfaces, and network functions that expand the attack surface. As a result, there is a growing emphasis on AI-driven and ML-based approaches to enhance anomaly detection and intrusion prevention capabilities in both the RAN and the 5G/B5G core [48].

Early efforts in this domain leverage ML techniques for attack detection at the network edge. For example, Xavier et al. [49] proposed an ML-based IDS xApp that utilizes physical and MAC layer features to detect DoS attacks before they reach the core network, while Amachaghi et al. [50] developed an ensemble of ML classifiers for anomaly detection. Building on these foundations, subsequent research has explored DL to enhance detection and mitigation capabilities. Kouchaki et al. [51] presented a dual xApp framework in which a self-attention-enhanced RNN autoencoder is used for anomaly detection, and

a Secure Slicing xApp orchestrates dynamic network slicing in response to detected threats, thereby integrating security and resource management to improve network resilience. Multilayer defense strategies have also gained attention. Soleymani et al. [52] proposed a framework that combines a dApp for rapid detection at the RAN level with xApps for in-depth analysis at the near-RT RIC, achieving flexible, layered protection against DDoS attacks. Addressing vulnerabilities specific to the O-RAN architecture, Hung et al. [53] proposed an anomaly detector targeting threats originating from the E2 interface and third-party xApps, employing state machine analysis and conformance checking to identify and mitigate unauthorized or malicious signaling. Finally, the distributed and disaggregated architecture of next-generation RANs has enabled federated learning as a promising approach for privacy-preserving and scalable IDS deployment. Attanayaka et al. [54] introduced a peer-to-peer federated learning framework for anomaly detection in O-RAN, enabling decentralized training and secure parameter aggregation. Rumesh et al. [55] further extended this concept by integrating hierarchical FL within a digital twin environment, allowing for safe and effective model training and validation across multiple network slices.

A variety of ML and DL-based defense mechanisms have been proposed to counter DDoS attacks in 5G and B5G core networks. Sheikhi et al. [56] introduced an unsupervised federated learning approach utilizing autoencoders, where each 5G core node independently trains a local model and a central server aggregates these models to improve detection performance. Similarly, SENTINEL [57] presented a self-protecting control plane framework that leverages AI to identify and isolate malicious users responsible for launching DDoS attacks. DL techniques have also been explored, for example, Hussain et al. [58] developed a CNN-based system for detecting compromised network cells under attack, while the authors in [59] proposed a framework using LSTM networks to identify DDoS incidents in UE-related network traffic. Recent work has further applied sequence modeling for improved threat detection. Pell et al. [60] used LSTM networks to detect PFCP signaling attacks in 5G networks with high accuracy. Similarly, Djaidja et al. [61] proposed attention-based RNNs for early intrusion detection by leveraging packet sequence data, reducing detection latency.

B. XAI for Intrusion Detection

Although AI-driven IDSs have demonstrated notable improvements in detection accuracy, their opaque decision-making processes can impede the interpretability and validation of alerts by security analysts, potentially reducing trust and hampering incident response times. To address these limitations, recent survey studies [14], [15], [62], [63] have explored the integration of XAI techniques with anomaly detection models in network security. These surveys collectively emphasize the growing interest in XAI approaches for enhancing interpretability, transparency, and decision support in AI-based IDSs.

Several recent studies have proposed explainable AI-based frameworks for anomaly detection in RANs. Barnard et al. [69] proposed a two-stage pipeline combining XGBoost-based

TABLE I
OVERVIEW OF SURVEY PAPERS ON IDS, XAI, AND LLM INTEGRATION

Ref.	Year	ML/DL	XAI	LLM	Focus
[33]	2021	✓	✗	✗	Review of ML/DL techniques for NIDS, coverage of datasets, evaluation metrics, and challenges.
[34]	2023	✓	✗	✗	Classification of signature, anomaly, and hybrid IDS, emphasis on dataset and methodology usage.
[35]	2025	✓	✗	✗	Review of DL-based IDS lifecycle, covering data management, attack investigation, methodologies, and datasets.
[37]	2023	✓	✗	✗	Review of adversarial attacks and defenses for DL-based NIDS.
[14]	2022	✓	✓	✗	Survey of XAI-IDSs, explainability of black/white-box models, and general X-IDS architecture.
[15]	2024	✓	✓	✗	Review of XAI methods for network traffic classification and prediction.
[62]	2025	✓	✓	✗	Survey of XAI methods for anomaly-based IDS in IoT.
[63]	2022	✓	✓	✓	Analysis of XAI solutions in 5G/B5G, highlighting challenges and future trends.
[64]	2025	✗	✗	✓	Survey of LLM fundamentals, techniques, deployment strategies, and applications tailored to telecommunications.
[65]	2025	✗	✗	✓	Review of Transformer and LLM-based IDS, applications and challenges.
[66]	2024	✗	✗	✓	Review of LLMs in cybersecurity, including applications, vulnerabilities, mitigation strategies, and datasets.
[67]	2025	✗	✗	✓	Analysis of LLM applicability on various cybersecurity tasks.
[68]	2024	✗	✓	✓	Review of LLM-based IDS methods, applications, challenges, and trends.

supervised intrusion detection with SHAP explanations feeding a deep autoencoder for anomaly detection. Basaran and Dressler [70] introduced XAIomaly, a semi-supervised Deep Contractive Autoencoder (DeepCAE) that integrates Generative AI and XAI to enhance scalability and interpretability while reducing the computational complexity of DL models. Similarly, Marantis et al. [71] proposed the Explainable Anomaly Prediction (XAP) framework, which employs an LSTM Encoder-Decoder combined with an Isolation Forest and utilizes SHAP-based feature attribution for interpretable detection of RAN anomalies. In addition, Sun et al. [72] developed SpotLight, a multi-stage explainable generative model aimed at detecting and localizing anomalies in large-scale Open RAN deployments. Additionally, Mahbooba et al. [73] explored the use of decision tree models to enhance trust management in IDS through explainability, focusing on interpretable rules extracted from benchmark datasets. Gaspar et al. [74] evaluated LIME and SHAP applicability on a multi-layer perceptron for IDS, demonstrating how these XAI methods improve interpretability and trustworthiness of black-box intrusion detection models. Furthermore, Wali et al. [75] proposed a reliable IDS combining explainable AI with random forest (RF) models to enhance resilience against adversarial attacks without retraining.

C. LLM Agents for Intrusion Detection, Mitigation and Explainability

Recently, the integration of LLMs and IDSs has attracted significant attention, with several comprehensive surveys highlighting their potential to enhance detection accuracy, scalability, and interpretability across a broad spectrum of cybersecurity applications [64], [65], [66], [67], [68], [76]. Table I provides an overview of recent surveys covering ML/DL-based IDSs, XAI-IDS integration, and the use of LLMs for detection, mitigation and natural language insights. Based on this foundation, recent research has demonstrated the effectiveness of LLM in code vulnerability detection [77] and intrusion detection, showing that models such as BERT can provide good performance in terms of accuracy, robustness, and even privacy preservation [78], [79]. In the context of RANs, Moore et al. [80] proposed the use of LLMs as detection mechanisms, illustrating their applicability

in intelligent network slicing and real-time security for O-RAN-enabled wireless deployments.

Beyond basic translation and summarization, recent advances in the reasoning capabilities of LLMs have unlocked new avenues for autonomous decision support in cybersecurity, including the automated proposal of mitigation strategies in response to detected threats [81]. A notable research trend is the exploration of LLMs to enhance the explainability of AI-driven security systems, aiming to overcome the limitations of traditional XAI techniques, which often generate complex or overly technical outputs that are challenging for non-expert analysts to interpret. Recent studies have demonstrated a range of LLM-powered frameworks designed to address this gap. For instance, Wen et al. [82] introduced 6G-XSec, an explainable edge security framework that integrates LLMs with DL-based anomaly detection, improving security telemetry in O-RAN environments. Likewise, Mavrepisa et al. [83] developed x-[p]AIIn, a GPT-based LLM that tailors XAI explanations for both technical users and non-experts, making model outputs clearer and more actionable. Similarly, Ali and Kostakos [84] proposed HuntGPT, a dashboard that combines ML-based anomaly detection, XAI techniques (SHAP, LIME), and a GPT-3.5 Turbo conversational agent to translate complex threat detections into human-readable insights, improving decision-making and trust in AI-driven security systems. Furthermore, Liu et al. [85] introduced an LLM-based framework that delivers accurate and interpretable anomaly detection along with comprehensive, human-readable explanations. Lastly, Jüttner et al. [86] presented ChatIDS, which leverages LLMs to translate technical IDS alerts into intuitive language and actionable guidance specifically for non-expert users, thereby enhancing the accessibility and usability of cybersecurity analytics.

Although prior work has explored explainable anomaly detection or the use of LLMs for generating human-readable insights, they typically address these components separately. Table II presents a comparison with recent related works. Our proposed approach uniquely integrates both through two dedicated applications: one for intrusion detection and another for post-hoc explainability and natural language interpretation. Although systems like HuntGPT employ a similar XAI-LLM pipeline, they are neither designed for nor deployable within

TABLE II
COMPARISON OF IDS, XAI, AND LLM INTEGRATION ACROSS RELATED WORK

Feature	[78]	[79]	[80]	[82]	[83]	[84]	[85]	[86]	This Work
Detection Method	LLM-based	LLM-based	LLM-based	AE/LSTM	N/A	Ran. Forest	LLM-based	Signature based	Seq. Class.
XAI Integration	✓	✓	✓	✓	✓	✓	✓	✓	✓
LLM for Natural Language Insights	✓	✓	✓	✓	✓	✓	✓	✓	✓
LLM for Mitigation Reasoning	✓	✓	✓	✓	✓	✓	✓	✓	✓
Domain	General	IoT/IIoT	Open RAN	Open RAN	General	General	General	General	Next-Gen RANs

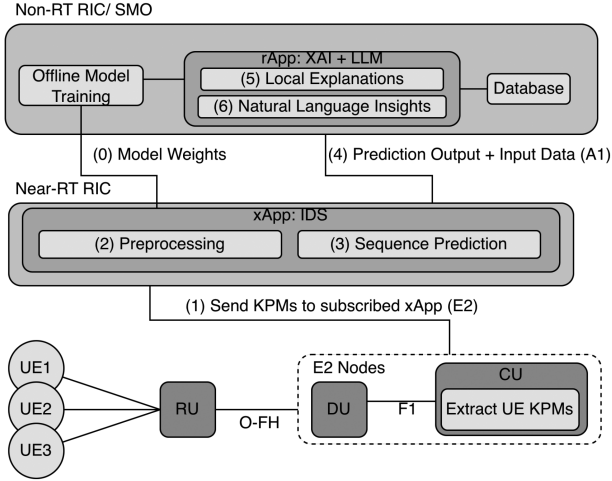


Fig. 1. End-to-end workflow of the proposed framework within the Open RAN architecture.

Open RAN environments. Our approach fills this gap by enabling near-real-time detection while providing interpretable, human-readable insights through LLMs. Unlike full LLM-based detection approaches that attempt to perform detection and interpretation within a single large model, our framework separates these functionalities into specialized modules. The decoupling of sequence classification and XAI-LLM improves deployment feasibility in latency-constrained AI-RAN environments.

III. INTERPRETABLE ANOMALY-BASED DDoS DETECTION SYSTEM

This section introduces our framework for intelligent DDoS detection and explainability within the Open RAN architecture. We integrate an LSTM-based detection model with XAI techniques and LLMs to provide interpretable insights. Section III-A details the architecture of the proposed framework and how it can be integrated into the Near-RT RIC. Section III-B focuses on the local explanation methods used to understand individual predictions, while Section III-C describes the LLM-based interpretability module.

A. Proposed Framework

The proposed framework is illustrated in Fig. 1. This framework integrates machine learning and explainability techniques within the Open RAN architecture to enable near real-time DDoS detection with interpretable insights. As traffic is exchanged from the UEs to the core network, the RAN components collect and extract KPMs for each UE. These KPMs are sent via

the E2 interface (Step 1) to the Near-RT RIC, where they are received by the IDS xApp. The framework consists of one xApp and one rApp:

- **xApp: IDS**, which includes a preprocessing module (Step 2) to organize KPMs into sequential input suitable for the classification model. The model performs DDoS prediction (Step 3), and both the prediction output and the corresponding preprocessed instance are forwarded to the XAI-LLM rApp (via the A1 interface) for further interpretation.
- **rApp: XAI + LLM**, which receives the output of the IDS xApp (Step 4). It first applies local explanation methods (SHAP and LIME) to interpret the prediction (Step 5), and then uses an LLM to translate these explanations into human-readable insights (Step 6). The decision to implement this functionality as an rApp is driven by both computational and architectural factors. Large-scale LLMs (e.g., GPT, DeepSeek) introduce notable inference latency, which can exceed the near-real-time constraints of xApps. rApps, being suited for non-real-time operations, are better aligned with the compute-intensive nature of XAI+LLM processing. Moreover, since this module serves an advisory role, providing interpretability and mitigation suggestions without directly interfacing with RAN components, its design naturally aligns with the rApp framework. Future extensions that automate mitigation could reconsider an xApp implementation.

The final output can be saved in a database for future use. Additionally, the Non-RT RIC includes an offline model training module, which periodically updates the model weights and deploys them to the xApp and rApp (Step 0).

B. XAI Methods

XAI addresses the challenge of providing transparency in the decision-making process of machine learning models. These techniques can generally be grouped into two major categories: intrinsic and post-hoc methods. Intrinsic approaches involve the use of inherently explainable models, such as tree-based or linear models. However, these may struggle or be unsuitable when working with high-dimensional or time-series data. In contrast, post-hoc methods aim to explain already-trained black-box models, such as neural networks, by analyzing their behavior after training. Post-hoc explainability techniques can be further divided into model-specific and model-agnostic approaches. The former are tailored to a particular model type and often leverage internal structures, such as attention weights, to provide insights. The latter, such as SHAP (SHapley Additive ex-Planations) [87] or LIME (Local Interpretable Model-Agnostic Explanations) [88], treat the model as a black box and generate explanations without relying on its internal structure, making

them applicable to a wide range of models. Finally, explanations can be categorized as local, which focus on individual predictions, or global, which aim to describe the overall behavior of the model.

1) *Local Explanations*: To interpret individual predictions made by the LSTM model, two local explainability methods were examined, LIME and SHAP.

LIME: constructs a local surrogate model around a specific prediction to approximate the behavior of a complex black-box model. The idea is to perturb the input sample multiple times and observe how these changes affect the model's predictions. These perturbed samples are weighted by their similarity to the original instance, and a simple, interpretable model, typically a sparse linear model, is trained to mimic the local decision boundary. LIME formalizes this process as the following optimization problem

$$\xi(x) = \arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g), \quad (1)$$

where f is the original black-box model, $g \in G$ is the family of interpretable models, $\mathcal{L}(f, g, \pi_x)$ measures how well g approximates f in the locality defined by π_x , and $\Omega(g)$ enforces simplicity in g (e.g., limiting the number of features). The solution $\xi(x)$ corresponds to the explanation for instance x . LIME is model-agnostic and flexible, making it useful for understanding individual predictions across a wide variety of models and data types [88].

SHAP: values are grounded in cooperative game theory and aim to fairly distribute a model's output among its input features. Each feature is treated as a "player" in a game, and its SHAP value reflects the average contribution it makes to the prediction across all possible feature subsets. The SHAP value for a feature i is defined as

$$\phi_i(f, x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \cdot [f_{S \cup \{i\}}(x) - f_S(x)], \quad (2)$$

where F is the full set of input features, $f_S(x)$ is the model's output when only the subset S is present, and $f_{S \cup \{i\}}(x)$ is the output when the feature i is added to the subset, representing the marginal contribution of feature i . SHAP values offer both local and global explainability and can be used with any model, although exact computation can be expensive for large feature sets. Computing exact Shapley values requires evaluating all 2^n feature subsets, which becomes unmanageable for high-dimensional inputs such as time-series data. To overcome this, Kernel SHAP was used, an efficient, model-agnostic approximation that fits a weighted linear model on randomly sampled subsets while preserving the core properties of Shapley values.

2) *Global Explanations*: In our paper, we used the global shapley values to provide a global explanation of our model. The global importance of a feature i can be estimated by aggregating its SHAP values across all samples in the dataset. This is typically done by taking the mean of the absolute SHAP values

$$\Phi_i = \frac{1}{N} \sum_{j=1}^N |\phi_i^{(j)}|, \quad (3)$$

where $\phi_i^{(j)}$ is the SHAP value of feature i for the j -th instance, and N is the total number of instances. The resulting value Φ_i reflects the average contribution (regardless of direction) of feature i to the model's predictions, offering a measure of global importance.

C. LLM for Human Interpretability

Outputs from XAI methods provide detailed insights into model decisions at the feature level, but they are often difficult to interpret or act upon, particularly for non-technical users. To bridge this gap, we introduce an LLM-based interpretability module designed to translate complex explanations into clear insights.

To enable consistent and context-aware interpretations, the input prompt provided to the LLM was carefully designed to encapsulate both global and instance-specific information, as detailed below. An example of such a prompt is shown in Fig. 2 for reference.

- *General Feature Statistics*: A table showing the mean and standard deviation of each feature under both normal and attack conditions, providing global context about the data distribution.
- *Model Input Sequence*: A matrix representing the input to the model, consisting of 3 time steps and 14 features, highlighting the temporal behavior of the instance being evaluated.
- *Model Output*: The binary classification result (normal or anomalous) produced by the classification model for the given input sequence.
- *Local Explanation Tables*: Tabulated outputs from LIME and SHAP explanation methods, showing the contribution of individual features to the specific prediction made by the model.
- *Global Feature Importance Table*: A table of mean absolute SHAP values computed across the dataset and structured by feature and time step, enabling the LLM to understand which features are generally most influential.
- *Task Instructions*: A final instruction block asking the LLM to produce a human-readable summary of the model's explanation and to suggest potential mitigation strategies. For LLMs that support reasoning, we explore whether enabling such modes improves performance on this task.

Initially, we adopted a zero-shot prompting strategy, in which the LLM was presented with the structured input prompt and asked to generate explanations without any prior examples. This approach enabled rapid evaluation of interpretability without the need for task-specific fine-tuning.

To further improve the quality of the generated insights, we applied a few-shot prompting strategy. Such strategies can provide the model with the necessary contextual cues through examples, while requiring relatively small datasets and low computational resources [89], [90]. In this approach, we appended the prompt with a couple example input-output pairs, each consisting of the previous prompt input followed by a high-quality, human-written explanation. The inclusion of these demonstrations can help the LLM better understand the structure, tone, and content expected in its responses. Fig. 3 provides the complete

Zero-Shot LLM Input Prompt

System Prompt: You are a cybersecurity expert analyzing LSTM outputs for DDoS detection. Respond with:

- **Anomaly Summary:** 3–6 bullet points
- **Misclassification Likelihood:** 1–2 bullets
- **Mitigation Steps:** 2–4 bullets

User Prompt:

General Data Distribution (Feature Statistics):

Feature	Normal Mean	Normal Std	Attack Mean	Attack Std
epre	-103.928	15.0925	-110.909	11.6904
pusch_snr	24.8102	8.7637	21.9557	6.3634
p_ue	-8.8639	11.6629	-9.7830	7.4086
ul_mcs	18.4491	7.5578	16.0374	7.4998
cqi	12.1369	3.1578	10.2456	3.7194
ul_bitrate (Mbps)	5.2858	8.3981	6.2719	6.8212
dl_mcs	19.2957	7.5593	15.7480	9.3631
dl_retx	34.6093	56.9998	41.6884	48.7054
ul_tx	604.935	688.756	779.463	382.341
dl_tx	589.642	800.250	288.121	157.916
ul_retx	56.1053	74.3320	107.043	86.4534
dl_bitrate (Mbps)	1.1481	1.5666	0.0456	0.0350
dl_err	0.0561	0.3432	0.3579	1.1688
ul_err	0.0985	0.5964	2.9725	10.8850

Input Sequence to LSTM Model (3 timesteps $\times$ 14 features):

Timestep	epre	pusch_snr	p_ue	ul_mcs	cqi	ul_bitrate	dl_mcs	dl_retx	ul_tx	dl_tx	ul_retx	dl_bitrate	dl_err	ul_err
T0	-127.8	20.3	-16	18.3	15	13.9507	26.7	3	952	459	84	0.09399	0	0
T1	-127.8	20.4	-6	18.6	15	14.0553	26.4	1	943	454	90	0.09181	0	0
T2	-127.4	20.7	-6	18.6	15	14.2386	26.6	0	951	461	86	0.09406	0	0

Model Prediction Output: 1 (Anomalous)

Local Explanation Table (LIME):

Feature	Contrib.	Feature	Contrib.	Feature	Contrib.
ul_bitrate_t-2 > 10.2148	0.10788	0.0056 < dl_bitrate_t-2 ≤ 3.20952	-0.10629	ul_err_t-1 ≤ 0	-0.09446
ul_bitrate_t-1 > 10.2148	0.08704	ul_err_t-2 ≤ 0	-0.05171	ul_err_t-0 ≤ 0	-0.04773
ul_bitrate_t-0 > 10.2148	0.04700	dl_err_t-0 < 0	0.03986	22 < dl_tx_t-2 ≤ 1401	0.03982
0.0056 < dl_bitrate_t-1 ≤ 3.20952	-0.03326	13 < cqi_t-2 ≤ 15	-0.03253	24 < ul_retx_t-0 ≤ 118.01	0.02780
-13 < p_ue_t-0 ≤ -1	-0.02673	19.7 < dl_mcs_t-1 ≤ 26.84	-0.02091	286 < ul_tx_t-2 ≤ 1289.86	0.01996
286 < ul_tx_t-1 ≤ 1289.86	0.01777	13 < cqi_t-0 ≤ 15	-0.01714	1.96 < dl_retx_t-2 ≤ 5	0.01650
-17 < p_ue_t-2 ≤ -13	0.01640	12.114 < ul_mcs_t-1 ≤ 18.738	-0.01485	epre_t-1 ≤ -113.752	0.01441
19.7 < dl_mcs_t-0 ≤ 26.84	-0.01296	-13 < p_ue_t-1 ≤ -1	-0.01255	18.488 < pusch_snr_t-2 ≤ 23.2	0.01130
22 < dl_tx_t-1 ≤ 1401	0.01066	epre_t-0 ≤ -113.752	-0.00906	24 < ul_retx_t-2 ≤ 118.01	-0.00891
dl_err_t-2 ≤ 0	0.00841	24 < ul_retx_t-1 ≤ 118.01	-0.00823	18.488 < pusch_snr_t-0 ≤ 23.2	0.00820
286 < ul_tx_t-0 ≤ 1289.86	-0.00783	19.7 < dl_mcs_t-2 ≤ 26.84	0.00586	epre_t-2 ≤ -113.752	0.00537
12.114 < ul_mcs_t-2 ≤ 18.738	-0.00514	22 < dl_tx_t-0 ≤ 1401	0.00508	0.0056 < dl_bitrate_t-0 ≤ 3.20952	-0.00465
12.114 < ul_mcs_t-0 ≤ 18.738	-0.00433	13 < cqi_t-1 ≤ 15	0.00223	18.488 < pusch_snr_t-1 ≤ 23.2	-0.00197
dl_retx_t-1 ≤ 1.96	-0.00124	dl_err_t-1 < 0	-0.00095	dl_retx_t-0 ≤ 1.96	0.00068

SHAP Local Heatmap:

Timestep	epre	pusch_snr	p_ue	ul_mcs	cqi	ul_bitrate	dl_mcs	dl_retx	ul_tx	dl_tx	ul_retx	dl_bitrate	dl_err	ul_err
T0	0.0045	0.00981	0.0139	0	0	0.12825	-0.00583	0.01925	0.0897	0.05718	0.06637	0.26096	0	0.00393
T1	-0.00555	-0.00776	0	0	0.00396	0.06754	-0.00874	0.00558	0.04449	0.0464	0.03202	0.05308	0	0
T2	0	0	0.00119	0.00093	0	0.03068	0	-0.01058	0.00999	0.02351	0.01385	0.04435	0	0.007

Global SHAP Feature Importance:

Timestep	epre	pusch_snr	p_ue	ul_mcs	cqi	ul_bitrate	dl_mcs	dl_retx	ul_tx	dl_tx	ul_retx	dl_bitrate	dl_err	ul_err
T0	0.00364	0.00590	0.01102	0.00396	0.00344	0.04322	0.00380	0.01208	0.03652	0.03523	0.02743	0.15316	0	0.00373
T1	0.00338	0.00359	0.00995	0.00306	0.00318	0.02245	0.00376	0.00501	0.02161	0.02823	0.01617	0.07941	0	0.00311
T2	0.00317	0.00370	0.00617	0.00349	0.00335	0.00974	0.00345	0.01257	0.00877	0.02194	0.00903	0.04387	0.00002	0.00316

Task for the LLM:

Provide a human-readable summary of the model's decision. Highlight the most influential features or patterns that contributed to the classification. Assess the likelihood of misclassification and suggest actionable mitigation strategies for the network operator.

Fig. 2. Zero-shot LLM input prompt.

set of example inputs and corresponding human-written outputs used during the few-shot prompting process.

D. Inference Overhead and Deployment Considerations

While LLMs enhance interpretability by translating complex XAI outputs into human-readable insights, they also introduce

inference/deployment costs that must be considered for practical adoption [91]. To avoid burdening the real-time pipeline, we placed the interpretability module in the non-RT RIC, where explanations can be generated asynchronously. Deployment can follow two main strategies: cloud-based APIs or self-hosted open-source models [92]. APIs provide straightforward access to advanced models but incur recurring usage fees, variable latency,

Few-Shot LLM Input Prompt

System Prompt: You are a cybersecurity expert analyzing LSTM outputs for DDoS detection. Respond with:

- **Anomaly Summary:** 3–6 bullet points
- **Misclassification Likelihood:** 1–2 bullets
- **Mitigation Steps:** 2–4 bullets

Shot 1

User Prompt:

General Data Distribution (Feature Statistics):

Feature	Normal Mean	Normal Std	Attack Mean	Attack Std
epre	-103.928	15.0925	-110.909	11.6904
pusch_snr	24.8102	8.7637	21.9557	6.3634
p_ue	-8.8639	11.6629	-9.7830	7.4086
ul_mcs	18.4491	7.5578	16.0374	7.4998
cqi	12.1369	3.1578	10.2456	3.7194
ul_bitrate	5.2858	8.3981	6.2719	6.8212
dl_mcs	19.2957	7.5593	15.7480	9.3631
dl_retx	34.6093	56.9998	41.6884	48.7054
ul_tx	604.935	688.756	779.463	382.341
dl_tx	589.642	800.250	288.121	157.916
ul_retx	56.1053	74.3320	107.043	86.4534
dl_bitrate	1.1481	1.5666	0.0456	0.0350
dl_err	0.0561	0.3432	0.3579	1.1688
ul_err	0.0985	0.5964	2.9725	10.8850

Input Sequence to LSTM Model (3 timesteps $\times$ 14 features):

Timestep	epre	pusch_snr	p_ue	ul_mcs	cqi	ul_bitrate	dl_mcs	dl_retx	ul_tx	dl_tx	ul_retx	dl_bitrate	dl_err	ul_err
T0	0.49356	1.51613	-0.1875	0.6087	0.0	-0.00791	-0.05000	-0.10526	-0.21445	-0.00145	-0.20168	-0.00013	0	0
T1	0.45923	1.39516	-0.1875	0.6087	0.2	-0.00787	-0.07857	0.05263	-0.21290	0.00000	-0.20168	0.00000	0	0
T2	0.41631	1.35484	-0.1875	0.6087	0.0	-0.00791	-0.01429	0.00000	-0.21445	-0.00073	-0.20168	-0.00004	0	0

Model Prediction Output: 0 (Normal)

LIME Local Explanation:

Feature	Contrib.	Feature	Contrib.	Feature	Contrib.
0.0056 < dl_bitrate_t-2 ≤ 0.0056	0.21917	0.0056 < dl_bitrate_t-1 ≤ 0.0056	0.14588	0.0056 < dl_bitrate_t-0 ≤ 0.0056	0.09185
dl_tx_t-2 ≤ 22	-0.08086	ul_err_t-2 ≤ 0	-0.05939	ul_err_t-0 ≤ 0	-0.05754
dl_tx_t-1 ≤ 22	-0.05388	p_ue_t-2 ≤ -17	0.05155	13 < cqi_t-2 ≤ 15	-0.03607
0.083207 < ul_bitrate_t-2 ≤ 10.2148	-0.03510	ul_err_t-1 ≤ 0	-0.03447	22 < dl_tx_t-0 ≤ 22	-0.03342
p_ue_t-1 ≤ -17	0.02391	13 < cqi_t-0 ≤ 15	-0.02296	24 < ul_retx_t-0 ≤ 118.01	-0.02130
dl_retx_t-0 ≤ 1.96	-0.02123	dl_mcs_t-2 > 26.84	-0.01959	0.083207 < ul_bitrate_t-1 ≤ 10.2148	-0.01919
12.114 < ul_mcs_t-0 ≤ 18.738	0.01613	dl_err_t-1 ≤ 0	0.01582	24 < ul_retx_t-2 ≤ 118.01	0.01551
epre_t-0 ≤ -113.752	-0.01530	p_ue_t-0 ≤ -17	0.01384	23.2 < pusch_snr_t-2 ≤ 30.888	0.01351
18.488 < pusch_snr_t-1 ≤ 23.2	0.01220	dl_retx_t-1 ≤ 1.96	0.01185	286 < ul_tx_t-0 ≤ 1289.86	-0.01153
dl_err_t-2 ≤ 0	-0.01020	dl_err_t-0 ≤ 0	-0.01013	12.114 < ul_mcs_t-2 ≤ 18.738	-0.01000
epre_t-1 ≤ -113.752	0.00954	24 < ul_retx_t-1 ≤ 118.01	0.00481	dl_retx_t-2 ≤ 1.96	0.00410
dl_mcs_t-1 > 26.84	-0.00298	dl_mcs_t-0 > 26.84	0.00224	18.488 < pusch_snr_t-0 ≤ 23.2	0.00186
13 < cqi_t-1 ≤ 15	-0.00146	286 < ul_tx_t-2 ≤ 1289.86	0.00135	286 < ul_tx_t-1 ≤ 1289.86	0.00077
12.114 < ul_mcs_t-1 ≤ 18.738	-0.00062	0.083207 < ul_bitrate_t-0 ≤ 10.2148	0.00054	epre_t-2 ≤ -113.752	-0.00001

SHAP Local Heatmap:

Timestep	epre	pusch_snr	p_ue	ul_mcs	cqi	ul_bitrate	dl_mcs	dl_retx	ul_tx	dl_tx	ul_retx	dl_bitrate	dl_err	ul_err
T0	0.00623	0.00272	0.01286	0.00139	-0.00274	-0.06505	-0.00204	-0.00263	-0.04795	-0.02709	-0.02715	0.15938	0	0
T1	-0.00244	-0.00324	0.01143	-0.00237	0.00337	-0.02430	-0.00237	-0.00670	-0.02017	-0.02912	-0.00820	0.08259	0	-0.00520
T2	-0.00429	0.00326	0.00864	0.00524	0.00191	-0.00648	0	-0.01405	-0.00932	-0.01182	-0.00840	0.03763	0	-0.00350

Global SHAP Feature Importance:

Timestep	epre	pusch_snr	p_ue	ul_mcs	cqi	ul_bitrate	dl_mcs	dl_retx	ul_tx	dl_tx	ul_retx	dl_bitrate	dl_err	ul_err
T0	0.00364	0.00590	0.01102	0.00396	0.00344	0.04322	0.00380	0.01208	0.03652	0.03523	0.02743	0.15316	0	0.00373
T1	0.00338	0.00359	0.00995	0.00306	0.00318	0.02245	0.00376	0.00501	0.02161	0.02823	0.01617	0.07941	0	0.00311
T2	0.00317	0.00370	0.00617	0.00349	0.00335	0.00974	0.00345	0.01257	0.00877	0.02194	0.00903	0.04387	0.00002	0.00316

Assistant Response:

Anomaly Summary.

- Download speed is abnormally low: the downlink bitrate (≈ 0.004 Mbps) is far below normal and even lower than attack averages.
- Very little downstream traffic: $dl_tx \approx 20\text{--}21$ suggests minimal downlink activity.
- Radio quality looks healthy: strong cqi , $pusch_snr$, and ul_mcs ; negligible retransmissions and errors.

Misclassification Likelihood.

- Moderate chance of a false negative: healthy radio plus minimal traffic could also align with stealthy/partial DDoS.

Mitigation Steps.

- Add a post-prediction rule: flag cases where $dl_bitrate < 0.01$ Mbps and $dl_tx < 50$ across multiple timesteps even if predicted normal.
- Cross-check with core-network telemetry (packet rates, tunnel sessions, server responses) to catch low-traffic attacks.

Fig. 3. Few-shot LLM input prompt.

and potential data privacy risks. In contrast, self-hosting offers control, predictable performance, and reduced long-term cost at scale, but requires substantial upfront investment in dedicated hardware and energy resources. For self-hosted models, cost can be reduced through compression techniques such as quantization, pruning, and knowledge distillation [93], as well as by

adopting lightweight fine-tuned LLMs. For API-based solutions, where direct model optimization is not possible, strategies such as prompt simplification, selective querying, and caching can help reduce overall cost [94]. To ground this discussion, Table III provides illustrative estimates of deployment costs for representative LLMs, linking parameter counts to their approximate

Shot 2

User Prompt:

General Data Distribution (Feature Statistics):

Feature	Normal Mean	Normal Std	Attack Mean	Attack Std
epr	-103.928	15.0925	-110.909	11.6904
pusch_snr	24.8102	8.7637	21.9557	6.3634
p_ue	-8.8639	11.6629	-9.7830	7.4086
ul_mcs	18.4491	7.5578	16.0374	7.4998
cqi	12.1369	3.1578	10.2456	3.7194
ul_bitrate	5.2858	8.3981	6.2719	6.8212
dl_mcs	19.2957	7.5593	15.7480	9.3631
dl_retx	34.6093	56.9998	41.6884	48.7054
ul_tx	604.935	688.756	779.463	382.341
dl_tx	589.642	800.250	288.121	157.916
ul_retx	56.1053	74.3320	107.043	86.4534
dl_bitrate	1.1481	1.5666	0.0456	0.0350
dl_err	0.0561	0.3432	0.3579	1.1688
ul_err	0.0985	0.5964	2.9725	10.8850

Input Sequence to LSTM Model (3 timesteps $\times$ 14 features):

TimeStep	epr	pusch_snr	p_ue	ul_mcs	cqi	ul_bitrate	dl_mcs	dl_retx	ul_tx	dl_tx	ul_retx	dl_bitrate	dl_err	ul_err
T0	-1.04292	-0.21774	0.4375	-0.02174	0.2	1.34885	0.50714	-0.10526	0.51127	0.31835	0.50420	0.02787	0	0
T1	-1.04292	-0.22581	0.4375	-0.01449	0.2	1.36354	0.50714	0.02632	0.51904	0.31545	0.54622	0.02758	0	0
T2	-1.06438	-0.22581	0.4375	-0.02174	0.4	1.34992	0.46429	-0.10526	0.51127	0.31472	0.50420	0.02694	0	0

Model Prediction Output: 1 (Anomalous)

Local Explanation Table (LIME):

Feature	Contrib.	Feature	Contrib.	Feature	Contrib.
ul_bitrate_t-2 > 10.2148	0.10788	0.0056 < dl_bitrate_t-2 ≤ 3.20952	-0.10629	ul_err_t-1 ≤ 0	-0.09446
ul_bitrate_t-1 > 10.2148	0.08704	ul_err_t-2 ≤ 0	-0.05171	ul_err_t-0 ≤ 0	-0.04773
ul_bitrate_t-0 > 10.2148	0.04700	dl_err_t-0 ≤ 0	0.03986	22 < dl_tx_t-2 ≤ 1401	0.03982
0.0056 < dl_bitrate_t-1 ≤ 3.20952	-0.03326	13 < cqi_t-2 ≤ 15	-0.03253	24 < ul_retx_t-0 ≤ 118.01	0.02780
-13 < p_ue_t-0 ≤ -1	-0.02673	19.7 < dl_mcs_t-1 ≤ 26.84	-0.02091	286 < ul_tx_t-2 ≤ 1289.86	0.01996
286 < ul_tx_t-1 ≤ 1289.86	0.01777	13 < cqi_t-0 ≤ 15	-0.01714	1.96 < dl_retx_t-2 ≤ 5	0.01650
-17 < p_ue_t-2 ≤ -13	0.01640	12.114 < ul_mcs_t-1 ≤ 18.738	-0.01485	epr_t-1 ≤ -113.752	0.01441
19.7 < dl_mcs_t-0 ≤ 26.84	-0.01296	-13 < p_ue_t-1 ≤ -1	-0.01255	18.488 < pusch_snr_t-2 ≤ 23.2	0.01130
22 < dl_tx_t-1 ≤ 1401	0.01066	epr_t-0 ≤ -113.752	-0.00906	24 < ul_retx_t-2 ≤ 118.01	-0.00891
dl_err_t-2 ≤ 0	0.00841	24 < ul_retx_t-1 ≤ 118.01	-0.00823	18.488 < pusch_snr_t-0 ≤ 23.2	0.00820
286 < ul_tx_t-0 ≤ 1289.86	-0.00783	19.7 < dl_mcs_t-2 ≤ 26.84	0.00586	epr_t-2 ≤ -113.752	0.00537
12.114 < ul_mcs_t-2 ≤ 18.738	-0.00514	22 < dl_tx_t-0 ≤ 1401	0.00508	0.0056 < dl_bitrate_t-0 ≤ 3.20952	-0.00465
12.114 < ul_mcs_t-0 ≤ 18.738	-0.00433	13 < cqi_t-1 ≤ 15	0.00223	18.488 < pusch_snr_t-1 ≤ 23.2	-0.00197
dl_retx_t-1 ≤ 1.96	-0.00124	dl_err_t-1 ≤ 0	-0.00095	dl_retx_t-0 ≤ 1.96	0.00068

SHAP Local Heatmap:

TimeStep	epr	pusch_snr	p_ue	ul_mcs	cqi	ul_bitrate	dl_mcs	dl_retx	ul_tx	dl_tx	ul_retx	dl_bitrate	dl_err	ul_err
T0	0.00450	0.00981	0.01390	0.00000	0.00000	0.12825	-0.00583	0.01925	0.08970	0.05718	0.06637	0.26096	0	0.00393
T1	-0.00555	-0.00776	0.00000	0.00000	0.00396	0.06754	-0.00874	0.00558	0.04449	0.04640	0.03202	0.05308	0	0.00000
T2	0.00000	0.00000	0.00119	0.00093	0.00000	0.03068	0.00000	-0.01058	0.00999	0.02351	0.01385	0.04435	0	0.00700

Global SHAP Feature Importance:

TimeStep	epr	pusch_snr	p_ue	ul_mcs	cqi	ul_bitrate	dl_mcs	dl_retx	ul_tx	dl_tx	ul_retx	dl_bitrate	dl_err	ul_err
T0	0.00364	0.00590	0.01102	0.00396	0.00344	0.04322	0.00380	0.01208	0.03652	0.03523	0.02743	0.15316	0	0.00373
T1	0.00338	0.00359	0.00995	0.00306	0.00318	0.02245	0.00376	0.00501	0.02161	0.02823	0.01617	0.07941	0	0.00311
T2	0.00317	0.00370	0.00617	0.00349	0.00335	0.00974	0.00345	0.01257	0.00877	0.02194	0.00903	0.04387	0.00002	0.00316

Assistant Response:

- **Anomaly Summary:** The model classified this sequence as **anomalous** due to persistent high uplink activity and low downlink usage, a pattern common in outbound DDoS behavior across the three timesteps.
 - ul_bitrate is consistently high (~1.35), prominently driving the decision (LIME: ul_bitrate_t-2/t-1/t-0 > thresholds with positive contributions).
 - Retransmission/erroneous metrics remain low-to-moderate, while traffic asymmetry (high UL vs. modest DL) is sustained.
 - LIME/SHAP highlight ul_bitrate, ul_tx, and constrained dl_bitrate as key factors.
- **Misclassification Likelihood:** Moderate but credible; heavy legitimate uploads (e.g., backups) could mimic this pattern, but multi-timestep persistence and low DL context strengthen the anomaly signal.
- **Mitigation Steps:**
 - Apply targeted UL rate-limits when high-UL/low-DL persists over N windows.
 - Validate with flow logs/PCAP (endpoints, ports, packet rates) to confirm intent.
 - Add symmetry features (UL/DL ratio, bitrate vs. CQI) and benign high-UL exemplars to training data.

Fig. 3. (Continued)

inference complexity (TFLOPs per token), API pricing, latency, and per-request costs under our zero-shot and few-shot settings.

IV. FRAMEWORK ANALYSIS

This section performs an evaluation of the proposed framework. Section IV-A describes the dataset and the pre-processing steps. Section IV-B details the simulation setup. Finally,

Section IV-C shows the numerical results of the classification model performance, XAI module explanations and finally the LLM interpretations.

A. Dataset

The dataset used in this study comprises raw 5G radio and core metrics obtained from the National Centre of Scientific Research

New Case

Actual Query User Prompt:

Normalized General Data Distribution (Feature Statistics):

Feature	Normal Mean	Normal Std	Attack Mean	Attack Std
epre	0.5695	0.2032	0.4760	0.1560
pusch_snr	0.6561	0.0997	0.6250	0.0732
p_ue	0.3921	0.2464	0.3714	0.1638
ul_mcs	0.6775	0.2775	0.5885	0.2767
cqi	0.8097	0.2172	0.6855	0.2649
ul_bitrate	0.0766	0.1215	0.0905	0.0983
dl_mcs	0.7158	0.2803	0.5847	0.3470
dl_retx	0.0455	0.0745	0.0541	0.0636
ul_tx	0.2864	0.3242	0.3674	0.1803
dl_tx	0.1623	0.2190	0.0790	0.0434
ul_retx	0.0864	0.1140	0.1630	0.1318
dl_bitrate	0.0162	0.0220	0.0006	0.0005
dl_err	0.0012	0.0085	0.0075	0.0262
ul_err	0.0008	0.0054	0.0233	0.0865

Input Sequence to LSTM Model (3 timesteps $\times$ 14 features):

Timestep	epre	pusch_snr	p_ue	ul_mcs	cqi	ul_bitrate	dl_mcs	dl_retx	ul_tx	dl_tx	ul_retx	dl_bitrate	dl_err	ul_err
T0	0.31930	0.37476	0.47826	0.66926	1.00000	0.42875	0.955	0.00909	0.45582	0.15981	0.340	0.00267	0	0
T1	0.30175	0.35564	0.45652	0.67315	0.93333	0.43123	0.965	0.00000	0.45630	0.15946	0.336	0.00268	0	0
T2	0.30175	0.36520	0.47826	0.66926	0.93333	0.43776	0.945	0.00260	0.46548	0.15981	0.356	0.00265	0	0

Model Prediction Output: 1 (Anomalous)

LIME Local Explanation:

Feature	Contrib.	Feature	Contrib.	Feature	Contrib.
ul_bitrate_t-0 > 0.22	0.09673	ul_retx_t-0 > 0.30	0.07885	ul_bitrate_t-2 > 0.22	-0.07646
0.72 < dl_mcs_t-1 ≤ 0.99	-0.06555	0.01 < dl_tx_t-0 ≤ 0.42	0.05546	ul_retx_t-1 > 0.30	0.05197
0.14 < ul_tx_t-0 ≤ 0.62	0.04699	epre_t-0 ≤ 0.37	0.04515	0.01 < dl_tx_t-1 ≤ 0.42	0.03832
ul_retx_t-2 > 0.30	0.03555	0.00 < dl_retx_t-2 ≤ 0.01	0.03123	0.14 < ul_tx_t-1 ≤ 0.62	0.02903
dl_err_t-1 ≤ 0.00	-0.02899	epre_t-2 ≤ 0.36	0.02518	ul_bitrate_t-1 > 0.22	-0.02365
0.14 < ul_tx_t-2 ≤ 0.62	0.02340	0.01 < dl_tx_t-2 ≤ 0.42	-0.02197	ul_err_t-0 ≤ 0.00	-0.02156
0.00 < dl_bitrate_t-0 ≤ 0.06	0.01579	0.31 < p_ue_t-2 ≤ 0.57	-0.01194	0.31 < p_ue_t-1 ≤ 0.57	-0.01120
epre_t-1 ≤ 0.36	0.01107	0.31 < p_ue_t-0 ≤ 0.57	0.01080	dl_err_t-2 ≤ 0.00	0.01077
dl_retx_t-1 ≤ 0.00	-0.01061	pusch_snr_t-1 ≤ 0.43	-0.01053	dl_retx_t-0 ≤ 0.00	-0.00867
ul_err_t-1 ≤ 0.00	-0.00809	0.45 < ul_mcs_t-0 ≤ 0.69	-0.00651	0.00 < dl_bitrate_t-2 ≤ 0.06	0.00641
0.87 < cqi_t-0 ≤ 1.00	-0.00577	pusch_snr_t-0 ≤ 0.43	-0.00525	0.00 < dl_bitrate_t-1 ≤ 0.06	0.00496
0.72 < dl_mcs_t-2 ≤ 0.99	-0.00479	0.45 < ul_mcs_t-1 ≤ 0.69	0.00422	0.87 < cqi_t-1 ≤ 1.00	-0.00378
0.45 < ul_mcs_t-2 ≤ 0.69	0.00343	dl_err_t-0 ≤ 0.00	-0.00265	0.72 < dl_mcs_t-0 ≤ 0.99	0.00181
ul_err_t-2 ≤ 0.00	-0.00100	pusch_snr_t-2 ≤ 0.43	-0.00078	0.87 < cqi_t-2 ≤ 1.00	0.00067

SHAP Local Heatmap:

Timestep	epre	pusch_snr	p_ue	ul_mcs	cqi	ul_bitrate	dl_mcs	dl_retx	ul_tx	dl_tx	ul_retx	dl_bitrate	dl_err	ul_err
T0	0.00271	0.00000	0.00000	0.00525	-0.00875	0.00000	-0.00923	0.00387	0.01334	0.06155	0.02720	0.06972	0.00000	0.00000
T1	0.00000	0.00320	-0.01210	0.01151	0.00000	0.03894	-0.01946	0.00784	0.04623	0.14729	0.03301	0.04863	0.00000	0.00000
T2	0.01356	0.00000	0.00592	0.00000	-0.01502	0.15978	-0.00992	0.01900	0.14422	0.16077	0.02937	0.02117	0.00000	-0.00522

Global SHAP Feature Importance:

Timestep	epre	pusch_snr	p_ue	ul_mcs	cqi	ul_bitrate	dl_mcs	dl_retx	ul_tx	dl_tx	ul_retx	dl_bitrate	dl_err	ul_err
T0	0.00975	0.00844	0.00853	0.01271	0.00466	0.01740	0.01768	0.01208	0.01597	0.09860	0.01823	0.03002	0.00051	0.00514
T1	0.00799	0.00981	0.01421	0.00769	0.00560	0.01767	0.03328	0.01418	0.02568	0.07379	0.02140	0.01703	0.00058	0.00429
T2	0.00549	0.00538	0.00705	0.00464	0.00482	0.04328	0.01549	0.01309	0.09734	0.06278	0.01995	0.01134	0.00053	0.00468

Task For LLM:

Provide a human-readable summary of the model's decision. Highlight which features or patterns most influenced the classification outcome. Assess the likelihood of misclassification and suggest actionable mitigation steps for the network operator.

Fig. 3. (Continued)

“Demokritos” [95]. This dataset was captured from a real-world 5G testbed following 3GPP standards. It documents various DDoS attack scenarios initiated by malicious UEs within a 5G network. The testbed consists of three cells connected to a shared 5G core, with nine UEs in total. Cells 1 and 3, along with the 5G core, are hosted on the Amarisoft Classic platform, while Cell 2 utilizes the Amarisoft Callbox Mini. Up to five malicious UEs were chosen to launch different flooding attacks, including SYN,

ICMP, UDP Fragmentation, DNS Flood, and GTP-U Flood, as detailed in Table IV. Attack data were recorded over four days, and KPMs were gathered using a data collector interfaced with the 5G network. For each UE, the KPMs include but not limited to downlink/uplink bit rates, Channel Quality Indicators (CQI), total transmitted bytes and transmission errors, sampled every five seconds. The final dataset consisted of 686,009 KPM reports, including 674,553 benign and 11,456 malicious samples,

TABLE III

ILLUSTRATIVE INFERENCE COST ESTIMATES FOR SELECTED LLMs. FLOPS PER TOKEN ARE APPROXIMATED BY $C \approx 2N$, WHERE N IS THE PARAMETER COUNT. VALUES ARE SHOWN IN TFLOPS PER GENERATED TOKEN, WITH INDICATIVE API PRICES, LATENCIES, AND PER-REQUEST COSTS.

Model	Params (N)	TFLOPS/token	Price in/1K	Price out/1K	API Latency/req.	Cost / request
GPT-4-Turbo [†]	$\sim 1.7 \times 10^{12}$	~ 3.4	\$0.01	\$0.03	~ 10 – 15 s	$\sim \$0.02$ (ZS), $\sim \$0.08$ (FS)
DeepSeek-V3-Chat	$\sim 6.71 \times 10^{11}$	~ 1.34	\$0.00056	\$0.00168	~ 13 – 20 s	$\sim \$0.002$ (ZS), $\sim \$0.005$ (FS)
Mistral-Large-Instruct-2411	$\sim 1.23 \times 10^{11}$	~ 0.25	\$0.002	\$0.006	~ 20 s	$\sim \$0.005$ (ZS), $\sim \$0.016$ (FS)

Notes: Estimates use the decoder-only rule of thumb $C \approx 2N$ FLOPS per token. Conversion: TFLOPS/token = $(2N)/10^{12}$. [†]GPT-4-Turbo parameter count is not publicly disclosed; value shown is an illustrative community estimate. Per-request costs assume ~ 1950 input tokens (zero-shot, ZS) and ~ 7000 input tokens (few-shot, FS), plus ~ 450 output tokens. Pricing sources: OpenAI (GPT-4-Turbo), DeepSeek (V3-Chat, cache-miss), Mistral (Large).

TABLE IV

OVERVIEW OF DDoS ATTACK TYPES, OCCURRENCE TIMES, AND PARTICIPATING DEVICES

#	Attack Type	Date	Time	Devices (IP)
1	SYN Flood	18.08.2024	07:00 - 08:00	10.20.10.2 10.20.10.4
2	ICMP Flood	19.08.2024	07:00 - 09:41	10.20.10.2 10.20.10.4
3	UDP Fragmentation	19.08.2024	17:00 - 18:00	10.20.10.2 10.20.10.4
4	DNS Flood	21.08.2024	12:00 - 13:00	10.20.10.2 10.20.10.4
5	GTP-U Flood	21.08.2024	17:00 - 18:00	10.20.10.2 10.20.10.4 10.20.10.6 10.20.10.8 10.20.10.10

TABLE V

COMPARISON OF FEATURES BETWEEN NORMAL AND ATTACK DURATIONS, SORTED BY ABSOLUTE DIFFERENCE

Feature	Normal		Attack		Diff. (%)
	Mean	Std	Mean	Std	
ul_err	0.10	0.69	2.94	10.90	2,906.13
dl_err	0.06	0.41	0.36	1.26	546.32
dl_bitrate (Mbps)	1.16	1.57	0.05	0.04	-96.05
ul_retx	56.67	74.79	106.92	86.46	88.67
dl_tx	592.88	799.94	288.56	158.50	-51.33
ul_tx	608.60	688.99	780.64	383.15	28.27
dl_retx	35.01	57.39	41.62	48.99	18.88
dl_mcs	19.33	7.57	15.79	9.37	-18.31
ul_bitrate (Mbps)	5.32	8.44	6.29	6.83	18.15
cqi	12.15	3.26	10.28	3.97	-15.39
ul_mcs	18.43	7.55	16.01	7.53	-13.13
p_ue	-8.78	12.07	-9.80	8.02	-11.62
pusch_snr	24.77	9.08	21.93	6.67	-11.47
epre	-103.97	15.14	-110.94	11.62	-6.70

reflecting a real-world imbalance where attack events are rare (only 1.7% of all samples). The attack samples were distributed as follows: SYN flood (1,402 samples, 12.2%), ICMP (3,756 samples, 32.8%), UDP fragmentation (1,402 samples, 12.2%), DNS flood (1,399 samples, 12.2%) and GTP-U flood (3,497 samples, 30.5%).

Exploratory Data Analysis: An Exploratory Data Analysis (EDA) was conducted to compare feature distributions during normal and attack periods. This analysis aimed to identify patterns, ranges, and potential indicators of malicious activity. Table V highlights the differences in mean values for key features, sorted by their percentage differences. For instance, uplink error rates (ul_err) and downlink error rates (dl_err) show a sharp increase during attacks, while downlink bitrates (dl_bitrate) drop significantly. These insights are supported

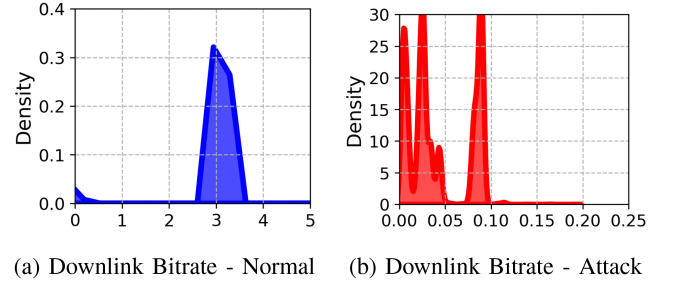


Fig. 4. KDE Comparison of Downlink Bitrate under Attack and Normal conditions. (a) Downlink Bitrate - Normal. (b) Downlink Bitrate - Attack.

by visualizations in Fig. 4, illustrating where data distributions become more skewed under attack conditions. Finally, Fig. 5 illustrates the temporal variation of some features across different attacks, comparing normal (blue) and malicious (red) UEs. The two rows represent different network parameters: uplink retransmissions (ul_retx), and uplink bitrate (ul_bitrate), while each column corresponds to a specific traffic type. The normal UE data were aligned with the same timestamps as the malicious UE to provide a direct, time-synchronized comparison. The distinct differences between normal and malicious traffic suggest that these patterns can help in more effective attack detection.

The dataset was further preprocessed to ensure high-quality input for the LSTM model. Features containing singular values or irrelevant identifiers (e.g., IP addresses) were excluded, while records with NaN values were removed. Data from periods of continuous transmission were selected to accurately represent UE behavior under both normal and attack scenarios.

B. Simulation Setup

We evaluated three deep learning architectures for sequential network traffic analysis: LSTM, TCN, and Transformer-based models. Each model was fine-tuned to identify the most effective hyperparameters, including the window size and the ratio of past data used. The dataset was partitioned into training and test sets using an 80–20 split, with RobustScaler normalization applied. All models were trained using the Adam optimizer with binary cross-entropy loss, a batch size of 64, and early stopping (patience = 5). The corresponding hyperparameter search spaces are summarized in Table VI.

Based on these results shown in Table VII, the LSTM achieved the best trade-off between accuracy, F1-scores, False Positive Rate (FPR), and False Negative Rate (FNR), inference time, and model size, therefore, it was selected as our final architecture.

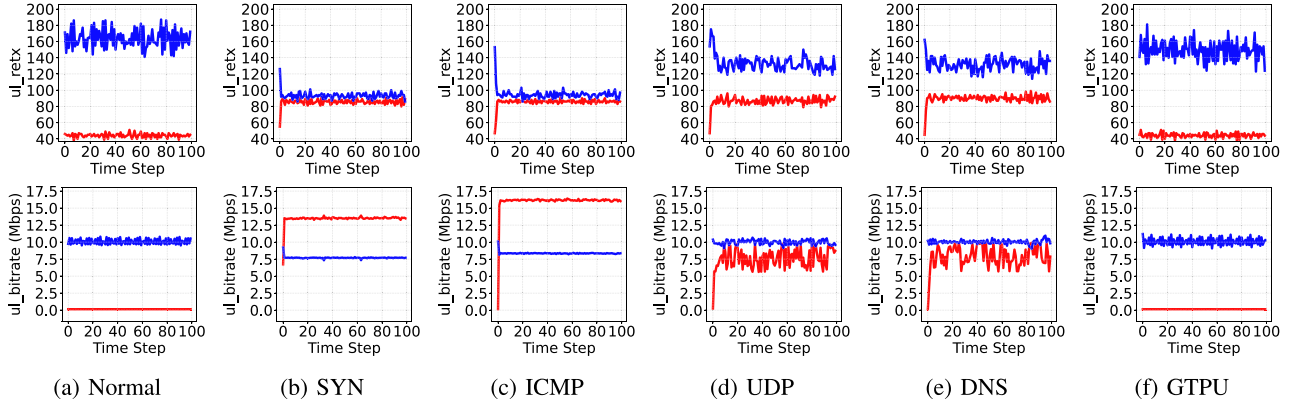


Fig. 5. Feature variation comparison between normal (blue line) and malicious (red line) UEs. Each column corresponds to a different network traffic type: Normal, SYN, ICMP, UDP, DNS, and GTPU, while each row represents a different network feature: ul_retx (row 1), and $ul_bitrate$ (row 2). The normal UE data is sampled at the same timestamps as the malicious UE for a direct comparison.

TABLE VI
HYPERPARAMETER SEARCH SPACES FOR EACH MODEL

Model	Hyperparameters (search range)
LSTM	$lstm_units \in \{16, 32, 64\}$, $num_layers \in \{1, 2\}$, $dropout \in \{0.1, 0.2\}$
TCN	$filters \in \{32, 64\}$, $kernel_size \in \{2, 3\}$, $dropout \in \{0.1, 0.2\}$, $num_blocks \in \{1, 2, 3\}$
Transformer	$head_size \in \{32, 64\}$, $num_heads \in \{2, 4\}$, $ff_dim \in \{64, 128\}$, $dropout \in \{0.1, 0.2\}$

We note, however, that while LSTMs were most effective in this setting, TCN and Transformer models may be more suitable in problems involving larger feature spaces or significantly longer temporal dependencies, where their ability to capture long-range patterns could provide advantages. The LSTM model architecture consists of an input layer processing sequences of shape (3,14), a 32-unit LSTM layer to extract temporal dependencies, and a dense output layer with sigmoid activation for binary classification.

Furthermore, we utilized multiple LLMs to generate human-readable summaries and mitigation strategies, selecting models of varying sizes to observe how output quality differs with scale. The models used, in descending order of approximate parameter count, were: *GPT-4-Turbo* (~ 1.7T parameters), *DeepSeek-V3-R1* (~671B), *Mistral-Large-Instruct-2411* (~123B), and Google's *Gemini-2.0-Flash* model. All models were accessed programmatically through their respective APIs. When supported, reasoning modes were enabled to assess whether such capabilities improved the quality of the generated outputs. We quantitatively assessed the outputs of the LLMs using standard readability metrics to evaluate their linguistic clarity and accessibility. Specifically, we used the *py-readability-metrics* library to compute two representative scores: (1) Flesch Reading Ease, where higher scores indicate easier-to-read text, and (2) Gunning Fog Index, which represents the U.S. school grade level needed to comprehend the text. In addition, we employed the *BERTScore* metric [96], which measures the semantic similarity between the LLM-generated outputs and human-written reference texts using contextual embeddings

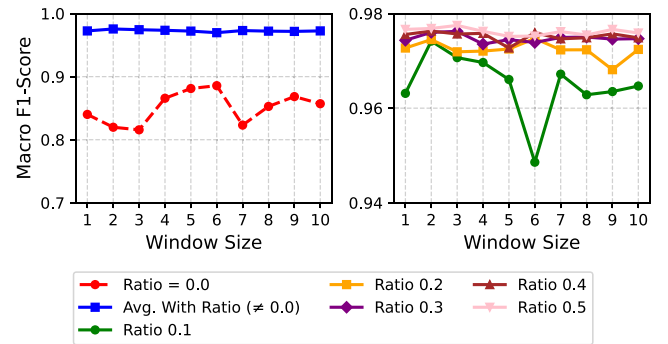


Fig. 6. Comparison of macro F1-score for no ratio vs. ratio.

from a pretrained transformer model (RoBERTa-Large). For each evaluation setup, 10 zero-shot and 10 few-shot prompts were crafted and executed twice per model, with the resulting metric values averaged to obtain consistent comparisons across models. Together, these metrics provide a comprehensive quantitative assessment of both the linguistic readability and semantic fidelity of the generated content across models.

C. Results

The first set of experiments evaluated the performance of the LSTM model in attack detection. Specifically, we examined how different combinations of window size and past data ratio influenced the overall model performance while addressing the challenge of catastrophic forgetting, where newly introduced training data overwrites previously acquired knowledge in sequential learning. Fig. 6 illustrates the effect of incorporating different ratios of past training data. The left subplot compares a scenario where no ratio is applied, resulting in a macro F1-score peaking at almost 0.90 on average, against one where past knowledge is retained (ratio $\neq 0$), maintaining macro F1-scores consistently above 0.95. The results indicate that incorporating a ratio stabilizes the model, preventing sharp fluctuations in performance. The right subplot further examines the impact

TABLE VII
COMPARISON OF BEST-PERFORMING CONFIGURATIONS FOR LSTM, TCN, AND TRANSFORMER MODELS

Model	Win. Size	Ratio	Acc.	Macro F1	Weighted F1	FPR (%)	FNR (%)	Inf. Time (ms\inst.)	Trainable Params
LSTM	3	0.5	0.9985	0.9775	0.9985	0.04	6.14	0.03	6,049
TCN	3	0.3	0.9985	0.9767	0.9985	0.05	6.20	0.04	7,681
Transformer	7	0.4	0.9984	0.9754	0.9984	0.05	6.54	0.09	18,999

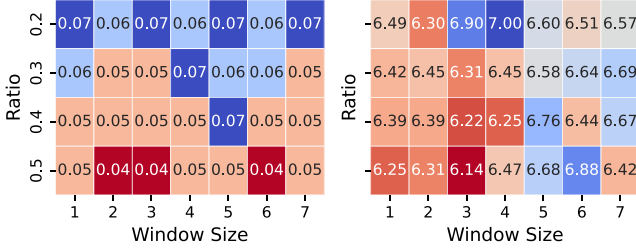


Fig. 7. Average FPR(%) (left) and FNR(%) (right).

TABLE VIII
AVERAGE F1-SCORE FOR PREVIOUS DAYS DATA WITH AND WITHOUT RATIO

Ratio	Day 1	Day 2	Day 3	Day 4
0.0	0.69	0.64	0.70	0.36
0.3	0.99	0.96	0.99	0.98

of various ratios ranging from 0.1 to 0.5, showing that while increasing the ratio improves performance, the gains become marginal beyond a certain threshold.

Beyond the effect of ratio selection, we also analyzed the impact of window size and ratio combinations on FPR and FNR, as shown in Fig. 7. The heatmap visualization highlights how these hyperparameters influence the trade-off between false alarms (left subplot) and missed detections (right subplot). The results indicate that different window size and ratio combinations lead to variations in FPR and FNR, though the overall differences remain moderate. Based on these findings, we selected a window size of 3 and a ratio of 0.3 as the optimal configuration for subsequent experiments, as they provided strong performance while keeping training times within a reasonable limit.

In addition, Table VIII presents the average macro F1-score of the final LSTM model for each previous day test set, comparing cases where no ratio (0.0) is used versus where a ratio of 0.3 is applied. The results clearly show that incorporating a ratio significantly enhances F1-scores across all previous day evaluations. Without a ratio, the model struggles to retain past knowledge, particularly for Day 4, where performance drops drastically to 0.36. In contrast, applying a ratio of 0.3 effectively preserves past knowledge, leading to consistently high macro F1-scores, all exceeding 0.96. These findings underscore the importance of balancing new and past training data to maintain model stability and performance, mitigating the risks of catastrophic forgetting.

Moreover, to address the class imbalance (only about 1.7% malicious samples), we employed a two-step resampling strategy: SMOTE [99] to oversample the minority class up to a target prevalence of roughly 10% (i.e., $\sim 1:9$ pos:neg), followed by Tomek links [100] to remove borderline samples. This primarily

TABLE IX
MISCLASSIFICATION BREAKDOWN PER ATTACK TYPE WITH AND WITHOUT SMOTE+TOMEK RESAMPLING

Attack	#Samples	Non-resampled (NR)	Resampled (R)
GTP-U	699	140 FN (20.0%)	66 FN (9.44%)
SYN	280	1 FN (0.36%)	2 FN (0.71%)
DNS	280	2 FN (0.71%)	1 FN (0.36%)
ICMP	751	0 FN (0.00%)	0 FN (0.00%)
UDP	280	0 FN (0.00%)	0 FN (0.00%)
Benign	134,891	61 FP (0.045%)	451 FP (0.334%)

TABLE X
PERFORMANCE COMPARISON WITH PREVIOUS WORKS USING NCSRD

Model	Weighted F1	Macro F1	FPR (%)	FNR (%)
kNN [97]	1.000	N/A	0.02	0.15
kNN [Ours] (NR)	0.996	0.950	0.07	14.10
kNN [Ours] (R)	0.995	0.931	0.32	9.25
XGBoost [97]	1.000	N/A	0.16	2.54
XGBoost [Ours] (NR)	0.997	0.961	0.19	4.27
XGBoost [Ours] (R)	0.997	0.965	0.15	4.76
CNN [98]	0.930	N/A	N/A	N/A
LSTM [98]	0.900	N/A	N/A	N/A
Proposed (NR)	0.998	0.977	0.04	6.24
Proposed (R)	0.996	0.947	0.33	3.01

reduces FNR at the expense of a higher FPR. Concretely, the non-resampled model achieved an overall FPR of 0.045% and FNR of 6.24%, whereas the resampled model achieved an FPR of 0.334% and an FNR of 3.01%. Table IX details FNs per attack family and FPs on benign traffic for both settings, showing that resampling markedly improves recall for the hardest class (GTP-U) while maintaining high precision overall.

Recent studies have evaluated the NCSRD dataset using a range of ML and DL models for DDoS detection. Christopoulou et al. [97] assessed classical ML algorithms such as k-Nearest Neighbors (kNN) and XGBoost, while Xylouris et al. [98] explored DL models including CNNs and LSTMs. Table X summarizes the performance of these approaches alongside our proposed method. To ensure a consistent and fair comparison, we re-implemented both ML baselines using 10 neighbors for kNN and representative XGBoost hyperparameters ($n_estimators=500$, $max_depth=6$). In our evaluation, neither model outperformed the proposed LSTM-based method. It is also worth noting that an F1-score of 1.00 corresponds to a theoretically perfect classification, implying zero false positives and zero false negatives—an outcome rarely achievable on real-world network traffic. Although the classical ML models achieved strong detection performance [97], their approach classified each instance independently, without leveraging temporal patterns across sequences, an essential factor for detecting evolving attack behavior. In contrast, our proposed DDoS detection captures temporal dependencies and achieves efficient inference, making it well-suited for near real-time deployment in AI-RAN environments. While Xylouris et al. [98]

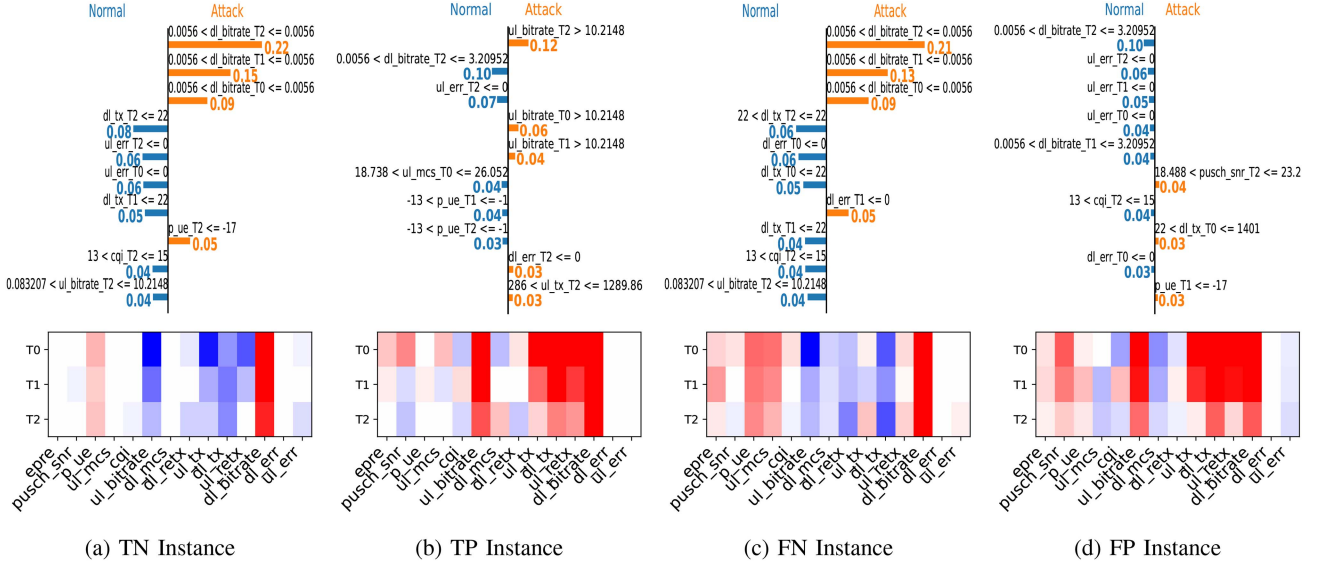


Fig. 8. Local explanations for TP, TN, FP, and FN instances. The top row shows the top 10 LIME feature contributions (feature name and value shown; blue: Normal, orange: Attack). The bottom row shows SHAP heatmaps over three time steps (T0-T2), where red supports "Attack" and blue supports "Normal".

also employed DL architectures capable of modeling temporal information (e.g., LSTM and CNN), differences in model architecture and feature selection contribute to the superior performance of our system, which operates at an average inference time of 0.03 ms per sample ($\sim 36K$ FLOPs) on a standard Intel i7-10700 CPU.

Having established the model's effectiveness, we now focus on explaining its decisions through local and global explanations, using LIME and SHAP to gain insights into individual and aggregated feature contributions. Fig. 8 presents local explanations for four representative test instances: a True Negative (TN), a True Positive (TP), a False Negative (FN), and a False Positive (FP). The top row shows the LIME-based explanations, highlighting the top 10 most influential features for each prediction. These explanations are based on a locally fitted linear model, which interprets each feature's contribution toward the predicted class (either "Normal" or "Attack"). LIME provides human-readable rules, for instance, it may indicate that a feature's value is greater than, less than, or falls within a specific range. As an example, in the second subplot of the top row, the feature `ul_bitrate` at time step T_2 contributes to an "Attack" prediction because its value is greater than 10.2 Mbps, as identified by the LIME model.

Furthermore, the second row of Fig. 8 provides local SHAP explanations of the four test instances. Each heatmap shows the SHAP values across the three time steps (T0-T2) and the 14 input features. Red regions indicate feature-time contributions that increased the likelihood of an "Attack" prediction, while blue regions represent contributions toward the "Normal" class. In the TP and FP instances, high-intensity red regions appear for features such as `ul_bitrate`, `ul_tx`, `ul_retx`, `dl_tx`, and `dl_bitrate`, indicating the model's strong reliance on these signals when predicting an attack. In the TN instance, although `dl_bitrate` shows strong attack-leaning contributions across timesteps, normal-leaning effects from `ul_bitrate`, `ul_tx`,

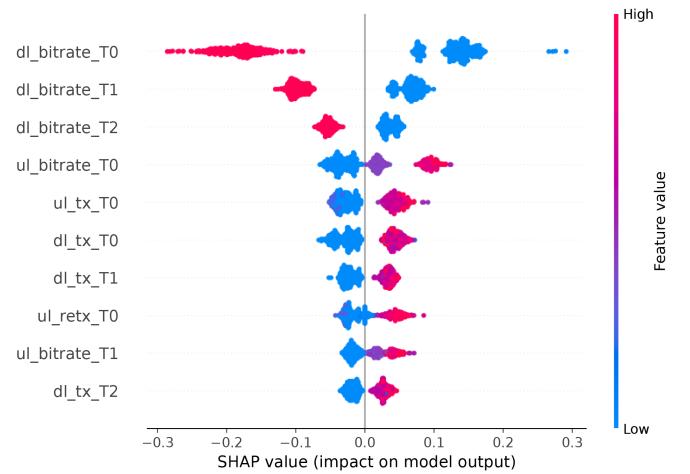


Fig. 9. Global SHAP beeswarm plot showing the top 10 most influential features across test instances.

and `dl_tx` counterbalance them, yielding a normal prediction. Finally, the FN instance exhibits a pattern similar to the TN case, except that other features collectively nudge the decision toward positive.

Furthermore, to understand which input features the LSTM model relies on globally when predicting DDoS, we leveraged the global SHAP explanations. Fig. 9 presents a beeswarm plot illustrating the top 10 most influential features based on SHAP value magnitudes. Each dot represents a SHAP value for a single instance, colors indicate the actual feature value (from low to high), while the x-axis shows the direction and strength of the feature's contribution. Features like `dl_bitrate`, `ul_bitrate_T0`, and `ul_tx_T1` emerge as highly impactful. To complement this, Fig. 10 provides a heatmap of the mean absolute SHAP values across test instances, aggregated per feature and time step. This gives a temporal perspective on

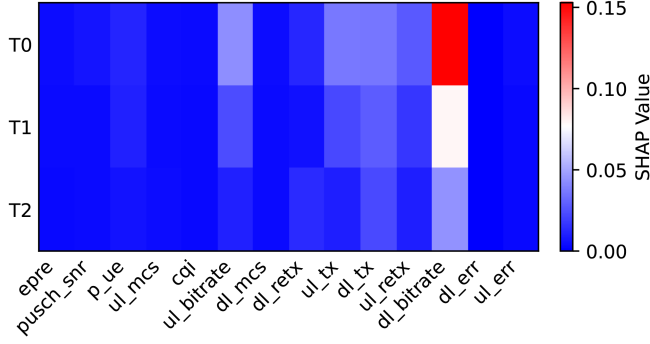
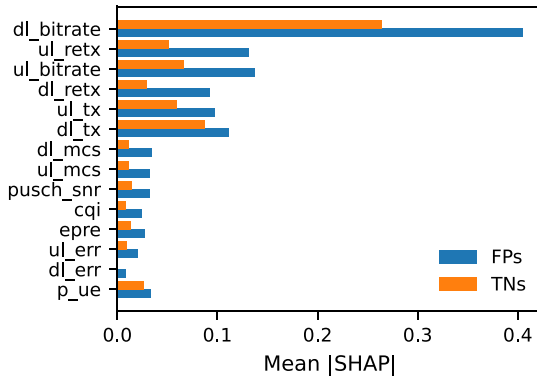
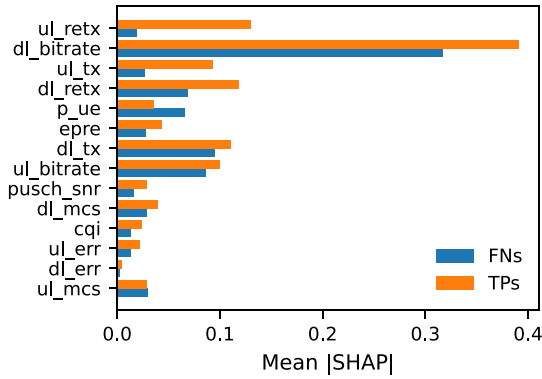


Fig. 10. Heatmap of the mean absolute SHAP values, showing the overall importance of each feature.



(a) FP vs TN



(b) FN vs TP

Fig. 11. Aggregate SHAP comparison between misclassified and correctly classified samples.

the importance of each feature. Notably, the most discriminative features cluster around time steps T_0 and T_1 , with metrics such as `dl_bitrate` to have the greatest contribution, and other notable features such as `ul_bitrate`, `dl_tx` and `ul_tx` showing relevance. These global explanation tools highlight the temporal and structural dependencies the model captures when identifying anomalous traffic patterns, which can potentially help in the final interpretations.

Finally, Fig. 11 contrasts the SHAP attributions of misclassified versus correctly classified samples, thereby uncovering

Anomaly Summary

- **High UL Bitrate:** The input sequence shows unusually high values for `ul_bitrate` across all timesteps, significantly deviating from the normal mean. This strongly indicates anomalous behavior, possibly due to a DDoS attempt to saturate uplink bandwidth.
- **Low DL Bitrate:** `dl_bitrate` values are extremely low compared to normal conditions, suggesting the downlink is being throttled or ignored in favor of abusive uplink traffic.
- **Stable Error Rates:** Both `dl_err` and `ul_err` are zero across all timesteps, which is unusual given other stressed network indicators.
- **High UL Retransmissions:** Elevated `ul_retx` implies repeated transmission attempts, a common DDoS characteristic where many packets are sent to disrupt service.
- **Consistently High UL TX:** Persistently high `ul_tx` further supports a DDoS pattern focused on the uplink channel.
- **Feature Contributions:** LIME shows `ul_bitrate` and `ul_retx` at multiple timesteps with strong positive contributions toward the anomalous label, reinforcing their importance in detection.

Misclassification Likelihood

- **Low likelihood of misclassification:** Given the large and consistent deviations in `ul_bitrate`, `ul_retx`, and related features across several timesteps, the model's anomalous decision aligns well with expected DDoS behavior.

Mitigation Steps

- **Traffic throttling:** Apply uplink rate limiting to curb excessive bandwidth use and mitigate impact.
- **Enhanced monitoring:** Increase visibility on the uplink to rapidly detect/respond to spikes or anomalous patterns.
- **Update filters:** Tune network filters to flag/block packets matching the observed DDoS traits (e.g., high retransmission rates, unusual bitrate patterns).
- **Incident response plan:** Activate DDoS playbooks: notify stakeholders, isolate affected systems, and engage security teams for deeper analysis and remediation.

Fig. 12. Zero-shot LLM output.

systematic patterns behind model mistakes. Subfigure (a) compares FPs against TNs, showing that benign flows misclassified as attacks receive disproportionately high attribution from features such as `dl_bitrate`, `ul_retx`, and `ul_bitrate`.

Subfigure (b) compares FNs against TPs, where attack flows that are missed exhibit consistently weaker contributions from key indicators such as `ul_retx` and `ul_tx`.

Having analyzed the model's behavior through local and global explanations, we transitioned to the final stage of the proposed framework, leveraging LLMs to transform these technical explanations into human-readable insights. As detailed in Section III-C, we designed special prompts for each test instance we wanted to interpret and then employed a zero-shot prompting strategy to generate informative summaries. The insights generated from a TP instance using zero-shot prompting are shown in Fig. 12.

The zero-shot prompting strategy used in this stage provided informative and context-aware insights that translated technical model outputs into language suitable for non-expert personnel.

TABLE XI
READABILITY AND SEMANTIC FIDELITY OF LLM-GENERATED EXPLANATIONS UNDER ZERO-SHOT AND FEW-SHOT PROMPTING

Model	Readability		Gunning Fog (grade) ↓	Fidelity (Human vs. LLM)		Consistency (LLM vs. LLM)	
	Flesch Reading Ease ↑			BERTScore-F1 ↑		BERTScore-F1 ↑	
	Zero-Shot	Few-Shot	Zero/Few-Shot	Zero-Shot	Few-Shot	Zero-Shot	Few-Shot
OpenAI	6.063	7.867	college graduate	0.793	0.800	0.858	0.854
OpenAI (reasoning)	33.027	27.224	college graduate	0.804	0.808	0.841	0.834
DeepSeek	14.874	13.091	college	0.799	0.808	0.876	0.875
DeepSeek (reasoning)	8.185	4.001	college graduate	0.794	0.806	0.846	0.854
Mistral	38.983	32.513	college	0.772	0.784	0.856	0.842
Gemma	29.825	27.025	college	0.786	0.810	0.828	0.838

Anomaly Summary

- **High Uplink Bitrate:** The model detected unusually high `ul_bitrate` across all timesteps versus the normal range, suggesting aggressive outbound traffic potentially consistent with DDoS participation.
- **Low Downlink Bitrate:** `dl_bitrate` stays consistently low, indicating the device is sending much more than it receives.
- **Stable Error Rates:** `ul_err` and `dl_err` are zero throughout, pointing to a stable link without integrity issues despite the traffic imbalance.
- **Consistent UL/DL Transactions:** `ul_tx` and `dl_tx` do not show spikes typically seen in bursty DDoS patterns; volumes are steady.

Misclassification Likelihood

- **Potential for False Positive:** Given stable errors and steady transaction counts, elevated `ul_bitrate` could stem from benign activities (e.g., backups, large uploads); the model may be sensitive to sustained high uplink.
- **Consider Contextual Information:** Incorporate application context, scheduled tasks, and user behavior to distinguish legitimate heavy uploads from attack-like traffic.

Mitigation Steps

- **Further Investigation:** Review flow logs/PCAPs to validate the nature of outbound data (endpoints, ports, payload characteristics).
- **Adjust Sensitivity:** If false positives recur, tune thresholds or feature weights related to `ul_bitrate` when errors and transaction dynamics appear normal.
- **User & Application Profiling:** Maintain baselines per user/app to contextualize high-throughput sessions and reduce false alarms.
- **Real-time Monitoring:** Deploy alerts for sudden pattern changes (e.g., surges in `ul_rettx`, shifts in UL/DL ratio) to complement LSTM outputs.

Fig. 13. Few-shot LLM output.

These summaries effectively described the rationale for the model's decision, highlighted key contributing characteristics. However, we also employed few-shot prompting to refine the LLM-generated outputs, enabling the insights to offer greater detail on the key drivers behind the model's classification. We presented the LLM with one TP and on TN sample with the appropriate outputs and then asked for the LLM to provide the human-readable insights for a third TP instance as shown in Fig. 13.

Lastly, we quantitatively assessed the outputs of the LLMs using standard readability metrics to evaluate their

linguistic clarity and accessibility. Specifically, we used the `py-readability-metrics` library to compute two representative scores: (1) Flesch Reading Ease, where higher scores indicate easier-to-read text, and (2) Gunning Fog Index, which represents the U.S. school grade level needed to comprehend the text. In addition, we employed the `BERTScore` metric [96], which measures the semantic similarity between the LLM-generated outputs and human-written reference texts using contextual embeddings from a pretrained transformer model (RoBERTa-Large). For each evaluation setup, 10 zero-shot and 10 few-shot prompts were crafted and executed twice per model, with the resulting metric values averaged to obtain consistent comparisons across models. Together, these metrics provide a comprehensive quantitative assessment of both the linguistic readability and semantic fidelity of the generated content across models.

Table XI presents the readability and semantic fidelity of explanations produced by different LLMs under zero-shot and few-shot prompting. Overall, models with explicit reasoning capabilities, achieve slightly higher semantic fidelity scores. In contrast, models like Mistral and Gemma produce more readable outputs, with simpler language and lower reading-grade levels. Across most models, few-shot prompting leads to small but consistent improvements in both readability and fidelity, suggesting that providing a few examples helps the models generate clearer and more faithful explanations. These findings highlight a trade-off between linguistic simplicity and semantic precision across different LLM families.

Finally, to evaluate the perceived usefulness of the generated explanations, we conducted a small-scale user study using the OpenAI reasoning model, which achieved the highest readability scores. Twelve telecommunication engineers assessed ten generated explanations (five benign and five attack) and rated each as clear/helpful or unclear/unhelpful. Participants rated 90% of benign explanations and 76.7% of anomaly explanations as clear and helpful, resulting in an overall positive rate of 83.3%, consistent with the quantitative readability and fidelity metrics achieved in Table XI.

V. CONCLUSION

This paper initially presented a survey of existing works on IDS, XAI, and LLM integration for RAN security, highlighting current research efforts and opportunities in this domain. We then introduced an LLM interpretable framework for detecting DDoS attacks in mobile networks, leveraging DL, XAI, and LLMs. Our results demonstrate that an LSTM-based model can

reliably detect anomalies in KPMs extracted from UEs with high accuracy. Additionally, integrating SHAP and LIME explanations with LLMs provides interpretable insights accessible to non-technical stakeholders. The inclusion of past data training ratio was also shown to be crucial in mitigating catastrophic forgetting, ensuring stable detection performance over time.

Future work will explore extending our framework to detect more sophisticated and stealthy attacks (e.g., low-rate or slow DDoS), as well as deploying the proposed xApp and rApp within a real Open RAN emulator environment to assess their operational performance and real-time integration capabilities. We also plan to automate the mitigation pipeline by translating LLM-generated insights into actionable control policies executable via the RIC, enabling a closed-loop detection and response mechanism. Lastly, to further improve the reasoning capabilities of the LLM, we will investigate the use of techniques such as Retrieval Augmented Generation (RAG), which can help the model generate more informed and context-aware mitigation strategies based on external information sources.

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